

NovaMart Retail — Project Documentation

Business case

NovaMart Retail is a fictional company used for a portfolio case study. The source is the public Sample Superstore dataset supplied by the user.

Source and scope

- 9,994 rows, 21 columns
- 2014-01-03 through 2017-12-30
- 5,009 distinct orders
- 793 customers
- 1,862 products
- 3 categories / 17 subcategories
- 4 US regions

Data quality

No missing values, full-row duplicates, invalid dates, date-order violations, non-positive quantities, negative sales or out-of-range discounts were found. Negative profit occurs on 1,871 lines and was retained because it is economically meaningful.

Cleaning

Dates were parsed. No rows were deleted. Repeated order/customer/product IDs were retained because the dataset is line-item grain. Outliers were identified descriptively but not removed automatically.

Metrics

Revenue = source Sales. Profit = source Profit. Margin = Profit / Revenue. Orders = distinct Order ID. AOV = Revenue / Orders. Repeat purchase rate = customers with ≥ 2 distinct orders / total customers.

Data distinction

Source fields remain identifiable. Calculated fields are explicitly labeled. Estimated Cost = Sales - Profit is derived from an accounting identity and is not a source cost. No simulated fields were added.

RFM

Customer recency, frequency and monetary value were derived from actual transaction history. Quintile-based scores were used to create descriptive segments.

Returns

No return fields exist in the supplied source. Return analysis is intentionally limited to a data-availability statement. No return rate or return reasons were fabricated.

Analytical standards

Discount analysis is descriptive and does not establish causality. Product and customer Pareto analysis quantifies concentration. Recommendations are framed as actions supported by observed patterns, not guaranteed outcomes.

Executive metrics

Revenue: \$2,297,200.86

Profit: \$286,397.02

Margin: 12.47%

Orders: 5,009

Customers: 793

Units: 37,873

AOV: \$458.61

Average discount: 15.62%

Repeat purchase rate: 98.49%

Key findings

- \$2,297,201 revenue across 5,009 orders and 793 customers.
- \$286,397 profit at 12.5% margin.
- Technology leads revenue.
- Technology leads total profit.
- Furniture has the lowest category margin.
- West leads revenue; Central has the weakest margin.
- Top 20 products represent 16.9% of revenue.
- Top 20 customers represent 11.5% of revenue.
- 306 products are loss-making in aggregate.
- 0% is the highest-profit observed discount band.
- Return KPIs are not calculable from the supplied source.

Priority recommendations

- Review loss-making products
- Tighten discount governance
- Monitor concentrated products/customers
- Regional margin diagnostic
- Target customer retention
- Integrate returns
- Experiment with promotions

Limitations

No unit cost, actual returns, store ID, or channel field is present in the supplied source. Therefore, analyses requiring those dimensions are not presented as source-backed findings.